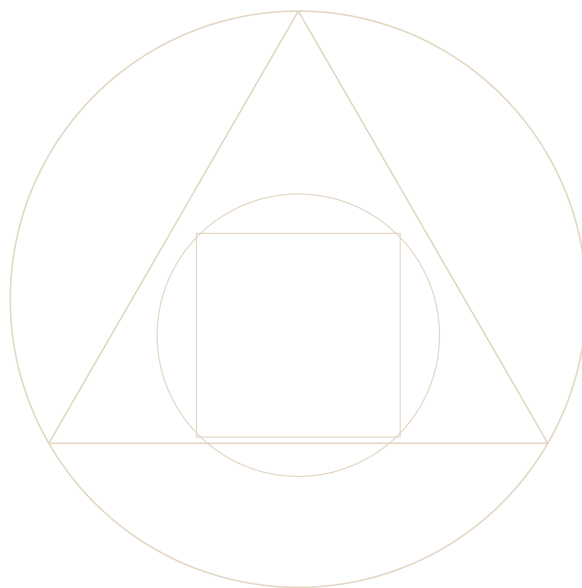




A COMPOUND REASONING SYSTEM ON ONE CONSUMER GPU

Eirenaeus-*Philalethes*

The same open model, used better: equal quality to the base model alone, measurably faster, with a verified memory.

1.51×

faster in total wall-clock time

677

problems across four public benchmarks

$p \geq 0.62$

no significant quality difference on any benchmark

12 GB

one consumer GPU, temperature 0, same harness

Selection over generation · Think as much as it pays · Measure before claiming

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BASE MODEL

Ternary-Bonsai-2-27B

ABSTRACT

The same weights, a better way to use them

Eirenaeus-Philalethes is a local compound AI system. It wraps an open 27-billion-parameter reasoning model with a small decision layer that runs on the CPU, a supervisor over the model's thinking, and a verified memory. It never changes the model's weights.

On 677 problems from four public benchmarks (HumanEval+, MBPP+, AIME 2025 and a held-out mix of code and math), measured on the same machine at temperature 0 against the same base model served alone, Eirenaeus shows **no statistically significant difference in quality** (McNemar exact test, $p \geq 0.62$ on every benchmark) while being **1.51× faster in total wall-clock time** (1.27× to 1.93× depending on the task). On tool calling (BFCL v4, 400 cases) it ties the base model in quality and gives no speed advantage; I report that too.

It pairs a fast model that decides with a capable model that reasons, a design others have also explored (Section 1). Everything here was measured on a single RTX GPU with 12 GB of memory. I report what worked, what did not, and the mistakes in my own measurement that I caught along the way, because a claim is only as good as the method behind it.

1.51×

faster overall

512 min vs 771 min

1.93×

best case

HumanEval+, 156 vs 154 solved

0

significant quality losses

across 677 paired problems

~0.5 s

decision on the CPU

before any GPU work starts

KEY FINDINGS

- 01 Same quality.** No benchmark shows a significant difference; on HumanEval+ Eirenaeus solves two more.
- 02 Faster where it matters.** The gain grows with how long the core would think: up to 1.93× on code.
- 03 No edge on tool calls.** Short calls need no thinking; the base model with thinking off is just as fast.
- 04 Method over luck.** One noisy run once suggested a loss that did not exist; every claim here is paired, at temperature 0.
- 05 Memory that finds the right thing.** On a frozen exam the exact memory reaches the model 14 times in 15, with no private leak.

CONTENTS

1	System 1 beside System 2	3	5	Results: tool calling	8
2	Motivation and architecture	4	6	Results: vision	9
3	Design principles and methodology	5	7	Memory, and what did not work	10
4	Results: reasoning	6	8	Limitations, roadmap and reproducibility	11

SECTION 1

System 1 beside System 2

Eirenaeus pairs a small, fast model that decides, a *System 1* in the sense of fast and slow thinking, with a capable reasoning model, a *System 2*. I arrived at this design independently, and others have explored the same pattern. This paper adds a packaged, local version and a paired measurement of what it costs and saves.

Related work. The closest system is the vLLM Semantic Router (Wang et al., 2025), where a ModernBERT classifier decides whether a served model should reason, reported at +10.2 points on MMLU-Pro with 47% less latency. Reasoning routers for hybrid models (Hugging Face, 2025) make the same call with a 0.6B model, and Laya's own authors present it as a System 1 for routing. Research pairs fast and slow parts in agents and planners (SwiftSage, 2023; System-1.x, 2024; Dualformer, 2024) and adapts thinking budgets per request (SelfBudgeter, 2025; DART and Ares, 2026).

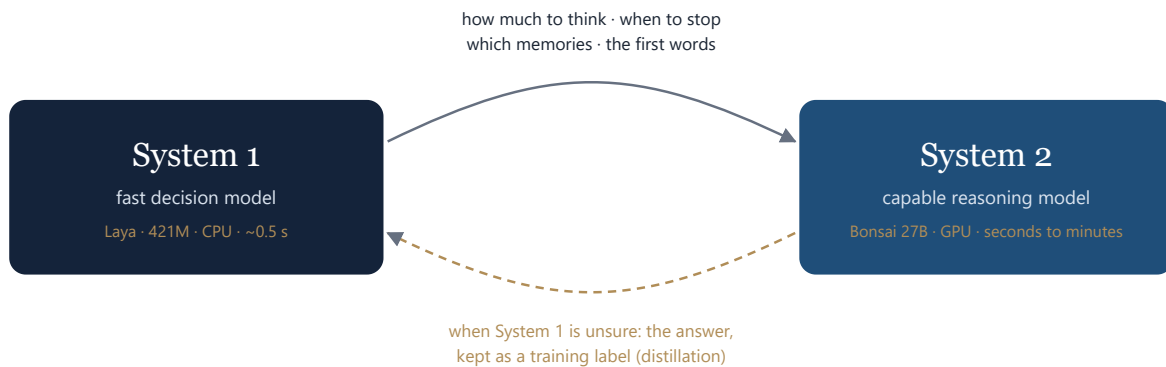


FIGURE 1 Two models, one loop. System 1 spends half a second on the CPU to decide how System 2 spends its GPU time. When System 1 is unsure, System 2 answers instead, and that answer becomes a lesson for System 1.

What the fast model gives the reasoning model

It decides how much to think before any GPU work, stops thinking that loops, picks the memories worth injecting and can open the reply with one line while the reasoning runs, for about half a second of CPU.

What the reasoning model gives the fast model

When the fast model is unsure, the reasoning model answers instead, and that answer is logged as a training label. Today the fast model is confident in 30% of routing decisions; the rest fall back to a safe rule.

WHAT SYSTEM 1 DECIDES TODAY	INSTEAD OF	COST
How much to think: none, brief, deep	always thinking at full budget	~0.5 s, CPU
Whether thinking is going in circles	running into the token cap	0 ms, a rule
Which three memories to inject	search calls made by the model	~0.45 s once

What this work adds. The pattern on one consumer GPU, packaged as a single model with an installer; a supervisor that stops looping thoughts; a verified memory (Section 7); the reasoning model's weights untouched; and a paired comparison at temperature 0 against the same model served alone (Section 6). Any capable open model can take the slow seat.

SECTION 2

Motivation

A strong open model on a consumer GPU spends most of its time thinking. Much of that thinking is either unnecessary, because the answer was clear early, or misdirected, because a simple request was treated as a hard one.

A system that decides *how much* to think, and notices *when* it can stop, should save time without losing answers. The question is narrow on purpose: **can an orchestration layer around a fixed model make it faster without making it worse?** The same weights, used better.

SECTION 3

Architecture

The parts are named after the three alchemical principles and the fifth essence. The names are a mnemonic, not a claim. A request reaches an OpenAI-compatible gateway; the decision layer reads it in about half a second and sends it down one of three paths.

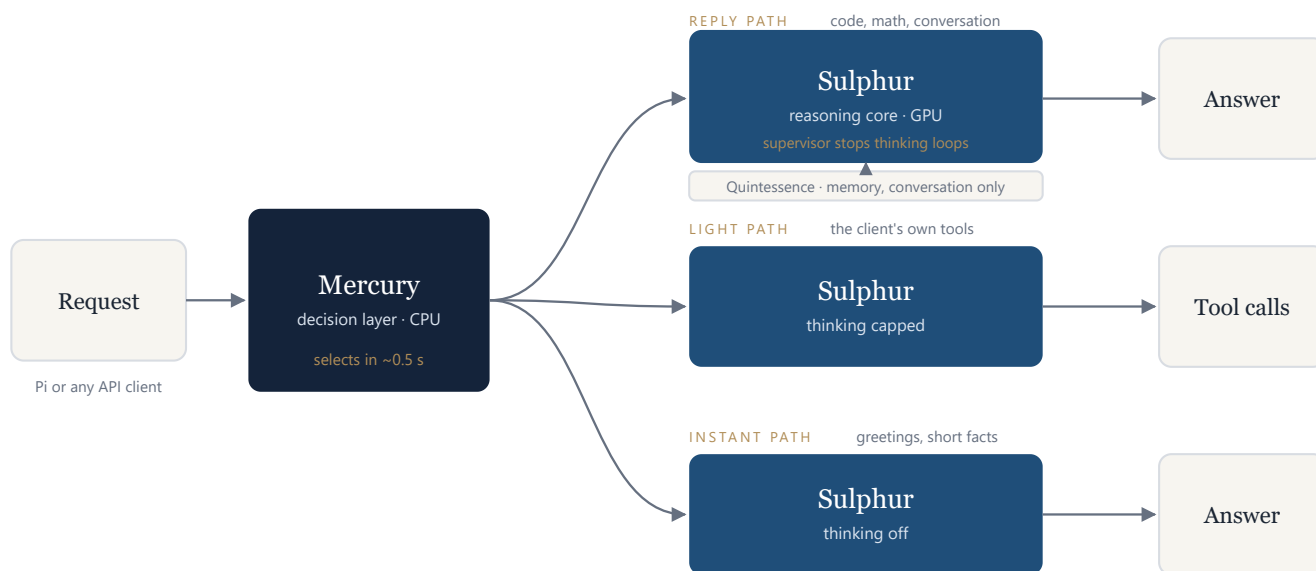


FIGURE 2 Request flow. The decision layer only selects; the reasoning core does the thinking, with as much effort as the decision layer chose.

PART	ROLE	RUNS ON
Mercury · decision layer	Laya, 421M parameters. Selects how much to think, whether the request is code, carries images, or needs live data.	CPU
Sulphur · reasoning core	Ternary-Bonsai-2-27B, PQ2_0, served by llama.cpp at about 65 tokens per second.	GPU · 10 GB
Salt · the body	One OpenAI-compatible endpoint, one model id, a local chat page.	CPU
Quintessence · memory	Reads Engram (Gentleman Programming); Dokimos judges what is still valid; three memories are injected, never searched by the model (Section 7).	CPU

Three paths. The *reply path* covers code, math and conversation: the core thinks with the effort the decision layer chose, and a supervisor watching the thinking stream stops loops (the same doubt repeated three times) by asking for the answer with what it has. Requests whose answer is the reply itself never receive memory and are never offered a tool they do not need. The *light path* serves clients that bring their own tools (a browser, a file system, an application): only the effort pick and a thinking cap. The *instant path* answers greetings and short facts without thinking.

SECTION 4

Design principles

I

Selection over generation

The decision layer never writes text; it chooses among options. Choosing is cheap, about 0.4 s on a CPU, and checkable; every escalation to the core becomes a training label.

II

Think as much as it pays

Thinking is the expensive part. A greeting gets none, a hard problem gets the whole budget, and a loop is stopped as soon as it repeats itself.

III

Measure before claiming

Every claim comes from a per-item results file: never from a console, never from one lucky run, and always against a control.

SECTION 5

Evaluation methodology

Same machine, same harness

Both arms answer the same questions with the same client and checker: Eirenaeus against the base model served alone with its authors' default settings.

Paired statistics

McNemar's exact test on the problems where the two arms disagree. "Better" or "worse" only below $p = 0.05$; otherwise, "no difference".

The control arm

When Eirenaeus seems to win through a setting, the base model is run with that setting too. If it matches, the credit goes to the setting.

Temperature 0

A single run with randomness is not evidence. Figure 2 shows a loss that was pure noise.

Held-out and judge splits

105 problems development never used; the memory exam keeps a frozen judge half, run once per milestone and never tuned on.

An unshared GPU

Every step logs memory use and any other GPU process; a shared GPU voids the run.

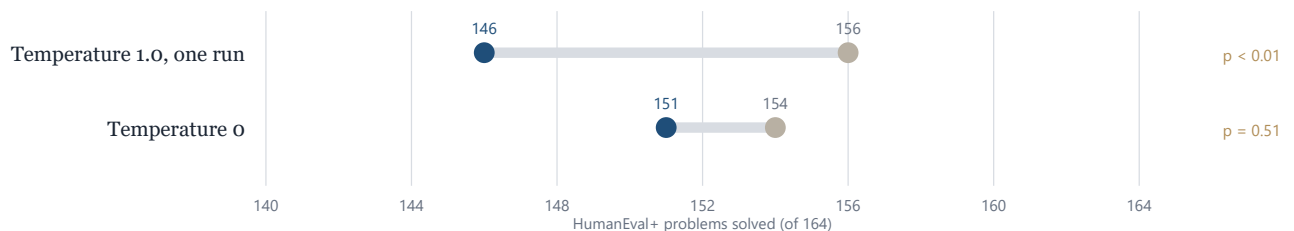


FIGURE 3 The noise lesson. One run at temperature 1.0 suggested Eirenaeus lost ten HumanEval+ problems ($p < 0.01$). At temperature 0 the gap disappeared ($p = 0.51$). I had already changed the system to fix a problem that did not exist.

SECTION 6

Results: reasoning

Across 677 paired problems, Eirenaeus solves as many as the base model alone and finishes the whole set in 512 minutes instead of 771: 1.51× faster, with no benchmark showing a significant difference in quality.

BENCHMARK	ITEMS	EIRENAEUS	BASE	ONLY E / ONLY BASE	MCNEMAR P	TIME E	TIME BASE	SPEED
Held-out mix	105	97	98	1 / 2	1.00	6,604 s	8,561 s	1.30×
MBPP+	378	308	309	7 / 8	1.00	12,227 s	19,228 s	1.57×
AIME 2025	30	27	27	1 / 1	1.00	6,827 s	8,691 s	1.27×
HumanEval+	164	156	154	3 / 1	0.62	5,055 s	9,752 s	1.93×
Total	677					30,712 s	46,231 s	1.51×

Temperature 0, one RTX GPU with 12 GB, Eirenaeus v2 against Ternary-Bonsai-2-27B served alone with its default thinking budget. "Only E / only base" counts the problems one arm solved and the other did not; those discordant pairs are what the McNemar test reads.

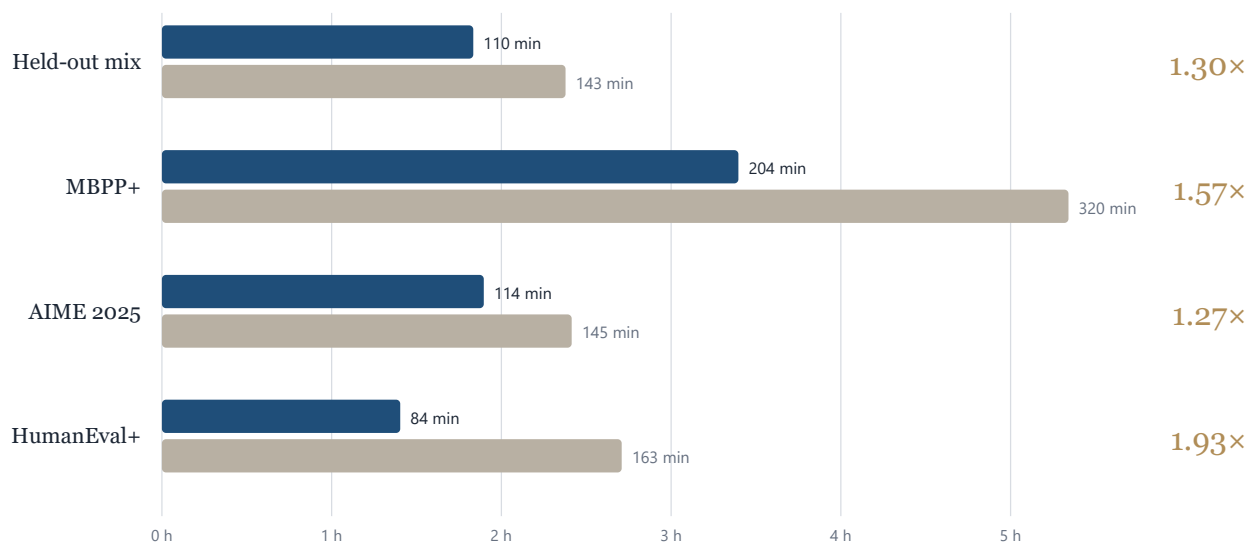


FIGURE 4 Wall-clock time per benchmark. Dark bars: Eirenaeus. Light bars: the base model alone. The gain is largest where the core would otherwise think the longest.

Where the time goes. On short problems Eirenaeus carries a fixed cost of one to two seconds per request (median 12.3 s against 11.2 s on the held-out mix). It wins that back many times over on problems where the core would think for minutes: the supervisor cuts loops, and code is no longer offered a tool it does not need.

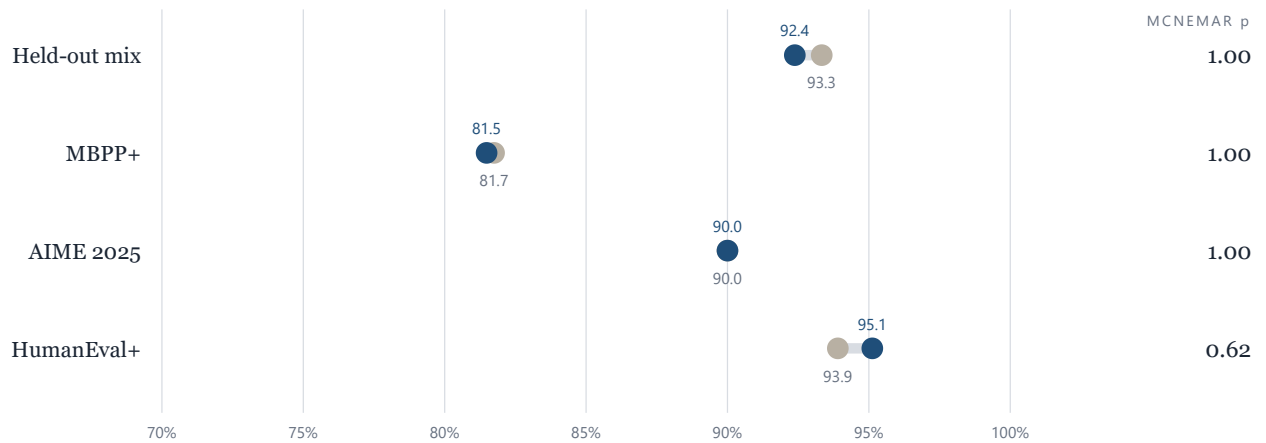


FIGURE 5 Accuracy per benchmark. Dark: Eirenaeus. Light: the base model alone. The right column is the McNemar p-value; every pair overlaps within noise.

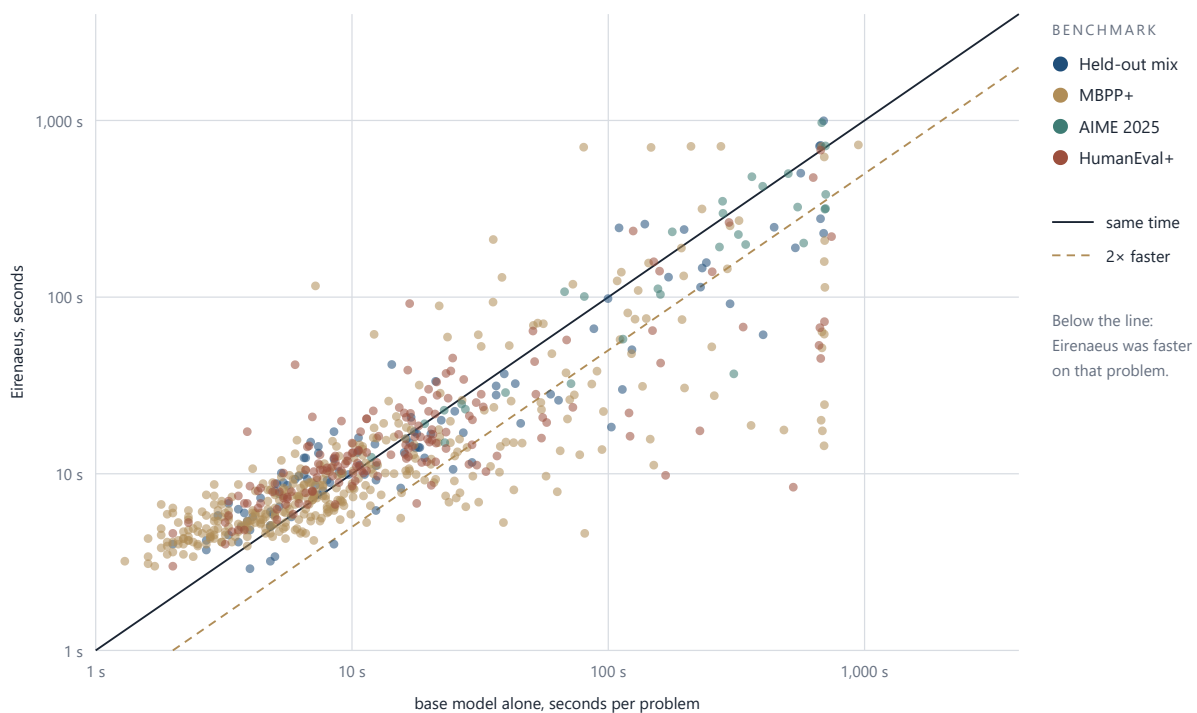


FIGURE 6 Every one of the 677 problems. Each dot compares the time the base model took (horizontal) with the time Eirenaeus took (vertical), on log scales. The cloud sits below the diagonal on long problems and just above it on very short ones.

What the speedup is

A property of the orchestration around one fixed model on one machine: deciding how much to think, stopping loops, and not offering tools a request does not need. The weights are identical.

What it is not

A faster model, or a result on other hardware. On very short requests the decision layer is pure overhead, which is why tool calling, next, shows no advantage.

SECTION 6.2

Results: tool calling

On BFCL v4 (simple, multiple, parallel and parallel-multiple calls, 100 each, official AST checker, temperature 0) Eirenaeus ties the base model in quality but brings no speed advantage. The control arm shows why.

CONFIGURATION	CORRECT	VS REFERENCE	TOTAL TIME	MEDIAN
Base alone, thinking (as published)	373 / 400	reference	1,589 s	3.1 s
Base alone, thinking off	371 / 400	$p = 0.84$	877 s	1.8 s
Eirenaeus, all layers	362 / 400	worse, $p = 0.04$	3,649 s	5.1 s
Eirenaeus, light path	375 / 400	tie, $p = 0.75$	2,027 s	3.8 s
Eirenaeus, light path, thinking off	368 / 400	$p = 0.42$	1,072 s	1.8 s

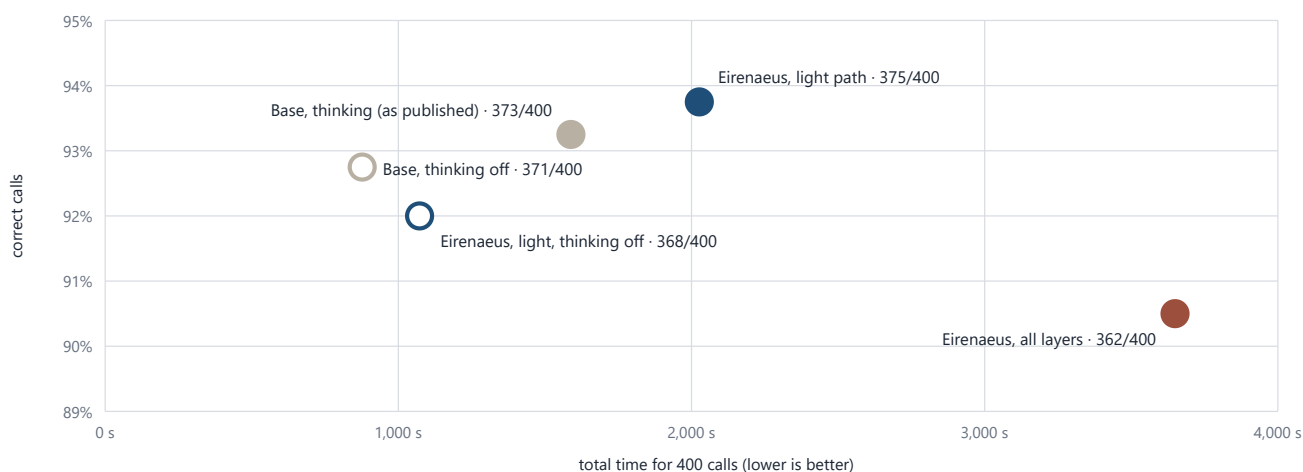


FIGURE 7 Accuracy against total time for 400 calls. Hollow markers: thinking off. Tool calls barely need thinking, so the fastest configuration is simply the base model with thinking off.

Reading the control. With every layer on, Eirenaeus was worse ($p = 0.04$): the date line in its prompt turned "this season" into the wrong year, and a live-data note leaked into unrelated calls. The light path restores quality (375 against 373). Turning thinking off makes everyone fast, the base model most of all, so that speed belongs to the setting, not to Eirenaeus. I report it as a tie.

Using a computer, through the client

Eirenaeus does not operate the computer itself. A client that brings its own tools, such as a browser automation, a file system or an application connector, keeps them: Eirenaeus decides when and how to call them on the light path above, and the client executes them with its own safeguards. The tie measured here is the quality of those calls.

SECTION 6.3

Results: vision, and the current version

Images were the one place where Eirenaeus lost clearly. The cause sat in the decision layer, not in the core: it read only the words of a request, so an exam question with a figure looked like a short factual answer and got no thinking at all.

BENCHMARK	ITEMS	EIRENÆUS	BASE	ONLY E / ONLY BASE	MCNEMAR P	TIME E	TIME BASE	SPEED
MMMU-Pro set 1 · v2	200	129	145	7 / 23	< 0.01	11,021 s	9,318 s	0.85×
MMMU-Pro set 2 · v3	200	130	137	9 / 16	0.23	7,256 s	7,848 s	1.08×
HumanEval+ · v3	164	156	154	3 / 1	0.62	5,014 s	9,752 s	1.94×
Held-out mix · v3	105	100	98	2 / 0	0.50	6,311 s	8,561 s	1.36×

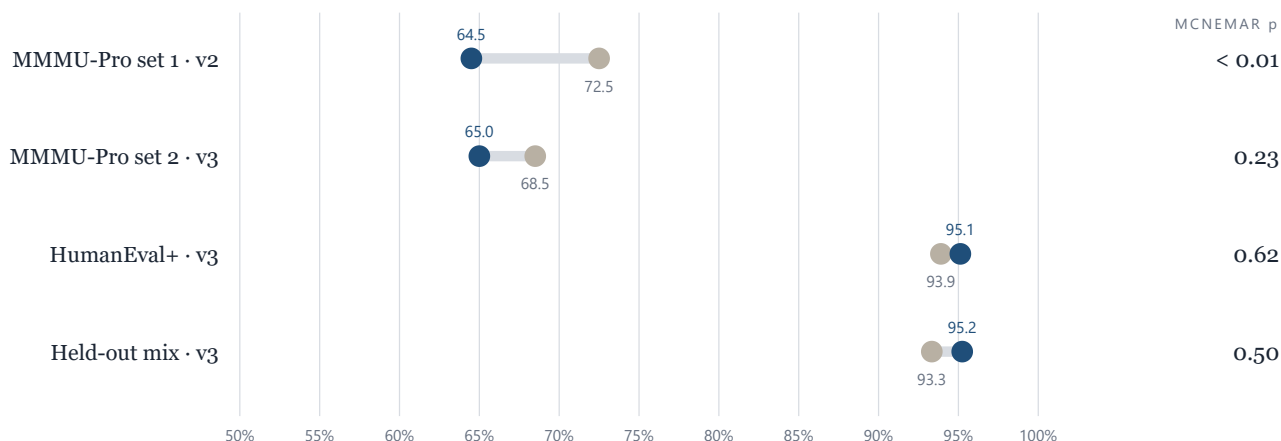


FIGURE 8 Accuracy, Eirenaeus (dark) against the base model alone (light). Set 1: 200 MMMU-Pro questions, v2. Set 2: 200 new questions never used in development, v3. Below them, the two reasoning benchmarks re-run with v3.

What changed in v3. The decision layer now receives a profile of the request (how many images it carries, whether tools are offered) and a request with images never goes to "no thinking". An exam answer with images also travels without Eirenaeus's own context. On set 1, v2 solved 16 fewer than the base model ($p < 0.01$, significant). On set 2, v3 solves 7 fewer ($p = 0.23$): no significant difference, and no advantage either.

A fix that broke the speed

The first v3 removed Eirenaeus's context from every direct answer, code included. HumanEval+ fell to 0.66× the base model's speed: one problem took 691 s instead of 22. That context is what keeps the core's thinking short. Limited to images, the speed came back (1.94×).

The current version

v3 was re-run on HumanEval+ and the held-out mix: same quality as the base model (156 vs 154, 100 vs 98) at 1.94× and 1.36×. MBPP+ and AIME 2025 were not re-run; the table in Section 6 is v2.

SECTION 7

Memory: fewer, fresher memories, chosen for free

Memory is written, judged and chosen by three different parts. **Engram** writes: each memory is structured (what, why, where, what was learned) and keyed by topic, so an evolving topic is updated instead of duplicated. **Dokimos** judges: every memory expires and is re-attested against its source, and an expired or replaced memory never reaches the model. **Eirenaeus** chooses: a privacy rule first, then the core rewrites the request as search keywords in both languages (about 0.45 s, once per conversation), then a word search on the CPU injects the top three memories, about 200 tokens. The model never spends a search call or a thought on remembering.

FROZEN MEMORY EXAM (AUTHOR'S WORK MEMORIES)	BEFORE	NOW
The exact memory that answers is on the card (judge half, 15 questions)	9 / 15	14 / 15
Small talk and private questions get no memory (incl. 8 fresh probes)	13 / 13	13 / 13
Expired memories on the card (without Dokimos: 2)	0	0
Memories eligible after privacy, project and Dokimos filters (of 4,508)	1,499	1,491
Cost of choosing the card	0.12 s CPU	0.45 s, once

The memories in this exam were written by the author's coding assistant, not by Eirenaeus; a memory Eirenaeus writes for itself is on the roadmap. Privacy is a rule over a user-defined list, never the model's judgement.

SECTION 8

What did not work

These are as informative as the wins, and each one changed the design.

Offering a tool the request does not need

Asked to write code, the model kept calling a tool to "test" it; each refusal cost a new round of thinking, and one problem took 47 minutes. Removing the offer removed the loop.

Fixing a problem that did not exist

A noisy single run suggested a code-quality loss; sending all code to maximum thinking cost the code speedup. The temperature-0 rerun showed there had been no loss (Figure 3).

Rewriting the conversation

A first "answer while thinking" put its opening line back into the conversation with an instruction to correct itself; asked for a square root, the model invented another calculation. The opening line is now a separate request, and the reasoning sees exactly the user's words.

Privacy left to the model

The core cannot know what is private to one person: when it wrote the memory keywords, a private topic reached the card once. Privacy became a fixed rule, checked on 8 fresh probes: 8 of 8.

SECTION 9

Limitations

- **One machine, one GPU class, one base model.** The speedup belongs to this setup until measured elsewhere.
- **Development touched public benchmarks.** The held-out mix and judge split guard against overfitting to them.
- **Short requests carry overhead.** The decision layer costs about half a second per request.
- **A small memory exam.** Thirty questions on one person's work memories; a public memory benchmark comes next.
- **No advantage on images.** On MMMU-Pro it is within noise of the base model (130 against 137).

SECTION 10

Roadmap

NEXT Verified early stop for code Stop thinking as soon as the code passes the examples in its own prompt; keep thinking only when it fails.	NEXT A memory it writes itself The decision layer flags what is worth keeping; the core writes it in the same structured form; Dokimos judges it.	THEN A public memory benchmark Long-term memory measured on conversations that are not the author's.	THEN Distillation Train System 1 on System 2's logged answers, so the fast model decides alone more often.
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SECTION 11

Reproducibility

Hardware	One NVIDIA RTX GPU with 12 GB of VRAM, a consumer desktop CPU, Windows 11.
Base model	Ternary-Bonsai-2-27B, PQ2_0, llama.cpp, 64K context, reasoning budget 36,864 tokens.
Decision layer	Laya, 421M parameters, on the CPU (revision 7b928d8 for the v3 runs, 55cf4c4 for the v2 table).
Benchmarks	HumanEval+ and MBPP+ (EvalPlus v0.2.0), AIME 2025, BFCL v4 with its official AST checker, and a held-out mix with a fixed offset that development never saw.
Evidence	Every number comes from a per-item results file (id, pass, seconds, route), available on request; every figure is generated from them.

HOW TO CITE

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Eirenaeus-Philalethes · J.C. - Asastu · Buenos Aires, October 2026. Set in Georgia and Segoe UI; figures generated from the result files at build time.

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